First ever
The 42-line Bible was the first book to be printed by Guttenberg with his movable type printing

More than ink
Printing technology has gone way beyond just inking text and images on flat sheets of paper

is another off-shoot that can construct large size models by sintering sand-like materials into moulds. Of all the exciting innovations in the field, the following need special mention:

Cornucopia

If you're really hungry and too tired to cook (or too broke to take-away) here, finally, is 'Cornucopia' - a printer that prints food. Scientist of the Fluid Interfaces Group Media Lab at MIT actually stores and refrigerates ingredients, mixes them and 'prints' the final product onto a serving tray. Though this is at the concept stage, it is an incredible innovation that's going to transform household spaces and

Digital gastronomy

family interaction patterns once it does hit the market. Amit Zoran and Marcel Coelho's machine's algorithm works by first selecting the relevant canisters of stored constituents on top of the machine, feeding them into a mixing chamber and then extruding them in layers of pre-set combinations. A touch-screen interface and internet connectivity augments the cooking process, and allows the user to keep track of the nutritive value and other such details of the recipe. Elementary, my dear Watson.

Builder's lobby

Enrico Dini – inventor and chairman of Monlite UK Ltd. Has created a massive 3D printer that can make entire buildings out of sand – and we're not talking about beach castles here. 'D-Shape' is a printer that sprays a layer of sand (about 5 to 10 mm each) and, through hundreds of little nozzles underneath, uses a magnesium-

Large scale printing with Dani's D-Shape

based binder to glue the sand into a hard, solid layer of stone, harder than even concrete. This process then works from bottom up to create an entire structure. The invention has been used to create a large, 9-cubic meter structure for a roundabout in Pontedera, Italy.

Second skin

Scientists and researchers at Wake Forest University in North Carolina have developed a bio-printer that that sprays skin cells on the wounds of a burn victim. The 'printer' is mounted on a frame and set to hover above the patient's bed. It then uses a laser to figure out the extend, width and depth of the burn and sprays out a layer of skin cells. This speeds up the recovery and helps the skin to grow back in about three weeks. Successfully tested on mice, it is an ideal alternative to other, more painful processes like skin grafting and chemical-based medication for burns.

Tissue culture

3D bio-printers can print more than just skin. Invetech (Australia) and Organovo (USA) have joined forces to construct organs, arteries and almost any body tissue by placing cells in a computer generated pattern using laser-calibrated print heads. The printer lets scientists arrange

Raise your glass to Vitraglyphic

cells of any kind in a 3D shape. It uses two heads – one for the cells and the other to set up support cells as a scaffold. The technology is likely to culminate in machines that could allow surgeons to make human body parts and organs whenever needed.

Glass printing

'Vitraglyphic' is a technique that makes glass objects from fine glass powder using CAD software and a 3D printer. The brain child of the Solheim Rapid Manufacturing Laboratory at the University of Washington, the printer uses inkjet printing technology to spread drops

Figure it out

of glass powder and fuse them using a binder solution. Glass isn't very good at absorbing water and so must be heated past its fusion point. This is ideal for architects, designers or artists who wish to build scale prototypes in glass.

Figuring it all out

It's no pipe dream that 3D printers of various types are likely to hit the world markets at affordable rate really soon. Which one of us can resist the ability to visualise an object and actually hold it in our hands within a few hours of the thought. It's amazing that a simple layering process could be developed to the extent that everything from tools to buildings, sculptures to subcutaneous tissues could be 'printed' out by machines in our own homes. Three-dimensional printers are a sign that the future is one of greater personalisation, greater power and perhaps even, the end of oligopolies and an equitable distribution of knowledge, resources and capital.